# Multi-Source Candidate Data Transformer — Design (Stage 1)

Turn messy, conflicting, multi-source candidate inputs into **one canonical profile** with provenance and confidence on every value. Principle: *wrong-but-confident is worse than honestly-empty* — unknowns become `null`, never invented.

## Pipeline

`ingest → extract (per-source adapters) → normalize → merge/dedupe → score confidence → canonical record → project (runtime config) → validate → emit`

**extract**: each adapter emits raw claims *(path, value, source, method)* and **never throws** — a bad source returns `[]` + a warning. **normalize**: pure functions canonicalize; a value that won't normalize is **dropped**. **merge**: claims → one CandidateProfile (single source of truth).

## Canonical schema & normalized formats

**Fields:** `candidate_id, full_name, emails[], phones[], location{city,region,country}, links{linkedin,github,portfolio,other[]}, headline, years_experience, skills[{name,confidence,sources[],verified_in_code}], experience[{company,title,start,end,summary}], education[{institution,degree,field,end_year}], provenance[{field,source,method}], overall_confidence`.

Phones → E.164; dates → `YYYY-MM` (a bare year is dropped, never invented as -01); country → ISO-3166 alpha-2 (exact or flagged-fuzzy); skills → canonical via an alias dictionary (js→JavaScript), unknowns kept & acronyms preserved; emails lowercased/validated.

## Sources (≥1 structured + ≥1 unstructured)

Structured: **Recruiter CSV**, **ATS JSON** (foreign field names → an **explicit remap dict**; unmapped keys logged, never guessed). Unstructured: **GitHub** (recorded fixture by default, `--live` for the real API), **Resume PDF/DOCX**, **Recruiter notes**.

## Merge & conflict resolution

**Identity / candidate_id — 3 tiers:** (a) hash of best email; (b) name + matching phone; (c) name + source-file identity. **Zero matchable identifiers → no cross-source dedupe** (kept standalone). **Scalars:** highest source trust wins (tie-broken by completeness, then stable order). **Lists** unioned & de-duped. Corroboration is case-insensitive. **Per skill `verified_in_code`:** each skill is `{name, confidence, sources[], verified_in_code}`; the flag is `true` iff `github` is among its sources — a skill claimed in a resume but absent from the candidate's public code carries this flag rather than being silently trusted as equally strong, and it does not alter confidence or add a scoring path.

## Confidence (concrete math)

| Source | ATS | CSV | Resume | GitHub | Notes |
|---|---|---|---|---|---|
| Base trust | 0.90 | 0.85 | 0.60 | 0.50 | 0.35 |

```
per_field = base_trust(winner) + 0.05 * min(corroborations, 3) # ≤ +0.15
per_field *= 0.9 if value is fuzzy/regex-derived (resume regex, fuzzy country)
per_field = clamp(per_field, 0.05, 0.99) # never 1.0
overall = importance-weighted mean (identity fields weighted higher)
```

**Worked examples (all reproducible from the shipped samples):** Jane `full_name` = ATS "Jane Mcdonald" (0.90) + CSV+GitHub+resume corroboration (+0.15) → clamp **0.99**; Liang `location.country` "The Netherlands"→NL is fuzzy, from ATS (0.90) + GitHub (+0.05 = 0.95) ×0.9 = **0.855**; Jane `PostgreSQL` skill, resume-only via regex (fuzzy) → 0.60×0.9 = **0.54**. *(Hypothetical, not in samples: a notes-only skill would floor at base 0.35.)*

## Runtime config (projection + validation)

Config selects/renames fields (`from` DSL: `emails[0]`, `skills[].name`), sets per-field normalization, toggles provenance/confidence, and chooses a missing-value policy driving **three schema shapes**: **omit** → not required; **null** → required + nullable; **error** → required + non-null (missing fails loudly). Canonical record and projection are strictly separated — default and any custom shape come from **one engine, no code changes**.

## Edge cases handled

(1) Conflicting name casing → trust winner kept, others still corroborate. (2) Malformed phone / bare-year date → dropped, stays null. (3) Malformed JSON / empty CSV → skipped with a warning; other sources still produce a profile. (4) Same skill, many spellings → canonicalized & merged, sources combined. (5) No email & no phone match → kept standalone; dedupe not attempted.

## Scope boundaries (intentional)

Chosen deliberately to keep the system deterministic and explainable: LinkedIn scraping is excluded (no public API); resume parsing uses deterministic regex/section heuristics rather than ML/NLP, and those values are flagged low-confidence; cross-candidate identity is limited to the 3-tier chain (no fuzzy clustering); and the surface is a clean CLI rather than a UI.